

Yosuke HIGUCHI

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Education

Waseda University

Ph.D. in Computer Science and Engineering

- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research topic: End-to-end Speech recognition

Tokyo, Japan

April 2021 - Present

Waseda University

M.E. in Computer Science and Engineering

- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research topic: End-to-end speech recognition

Tokyo, Japan

April 2019 - March 2021

Waseda University

B.E. in Computer Science and Engineering

- Supervisor: Tetsunori Kobayashi, Tetsuji Ogawa
- Research topic: Zero-resource speech recognition

Tokyo, Japan

April 2015 - March 2019

Work Experience

Google

Mentor: Andrew Rosenberg, Bhuvana Ramabhadran

- Research internship
- Worked on text-injection for speech recognition training

New York, NY, USA

October 2022 - January 2023

Carnegie Mellon University

Mentor: Shinji Watanabe

- Research Internship / Visiting Scholar
- Worked on integrating BERT into end-to-end speech recognition [Higuchi+, EMNLP2022]

Pittsburgh, PA, USA

March 2022 - Present

Mitsubishi Electric Research Laboratories

Mentor: Takaaki Hori, Niko Moritz, Jonathan Le Roux

- Research Internship / Part-time researcher (remote)
- Worked on semi-supervised end-to-end speech recognition [Higuchi+, Interspeech2021], [Higuchi+, ICASSP2022], [Higuchi+, JSTSP2022]

Cambridge, MA, USA

October 2019 - December 2021

Johns Hopkins University

Mentor: Shinji Watanabe

- Research Internship / Visiting Scholar
- Worked on non-autoregressive end-to-end speech recognition [Higuchi+, Interspeech2020], [Higuchi+, ICASSP2021]

Baltimore, MD, USA

December 2019 - March 2020

IBM Research AI

Mentor: Masayuki Suzuki, Gakuto Kurata

- Research Internship
- Worked on speaker diarization [Higuchi+, ICASSP2020]

Tokyo, Japan

June 2019 - October 2019

NTT Communication Science Laboratories, NTT Corporation

Mentor: Hirokazu Kameoka

- Research Internship
- Worked on voice conversion

Kanagawa, Japan

August 2018 - September 2018

Publication

Journal

- **Yosuke Higuchi**, Niko Moritz, Jonathan Le Roux, Takaaki Hori, “Momentum Pseudo-Labeling: Semi-Supervised ASR with Continuously Improving Pseudo-Labels,” *IEEE Journal of Selected Topics in Signal Processing (JSTSP)*, October 2022.

International Conference

- Yifan Peng*, Siddhant Arora*, **Yosuke Higuchi**, Yushi Ueda, Sujay Kumar, Karthik Ganesan, Siddharth Dalmia, Xuankai Chang, Shinji Watanabe, “A Study on the Integration of Pre-Trained SSL, ASR, LM and SLU Models for Spoken Language Understanding,” In *2022 IEEE Spoken Language Technology Workshop (SLT)*, January 2023.
- **Yosuke Higuchi**, Brian Yan, Siddhant Arora, Tetsuji Ogawa, Tetsunori Kobayashi, Shinji Watanabe, “BERT Meets CTC: New Formulation of End-to-End Speech Recognition with Pre-trained Masked Language Model,” In *Findings of the Association for Computational Linguistics: EMNLP 2022*, December 2022.
- Masao Someki, **Yosuke Higuchi**, Tomoki Hayashi, Shinji Watanabe, “ESPnet-ONNX: Bridging a Gap Between Research and Production,” In *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference 2022 (APSIPA)*, November 2022.
- Keqi Deng*, Zehui Yang*, Shinji Watanabe, **Yosuke Higuchi**, Gaofeng Cheng, and Pengyuan Zhang, “Improving Non-Autoregressive End-to-End Speech Recognition with Pre-trained Acoustic and Language Models,” In *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.
- **Yosuke Higuchi**, Niko Moritz, Jonathan Le Roux, Takaaki Hori, “Advancing Momentum Pseudo-Labeling with Conformer and Initialization Strategy,” In *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.
- **Yosuke Higuchi**, Keita Karube, Tetsuji Ogawa, Tetsunori Kobayashi, “Hierarchical Conditional End-to-End ASR with CTC and Multi-Granular Subword Units,” In *2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2022.
- Huaibo Zhao, **Yosuke Higuchi**, Tetsuji Ogawa, Tetsunori Kobayashi, “An Investigation of Enhancing CTC Model for Triggered Attention-based Streaming ASR,” In *Proceedings of Asia-Pacific Signal and Information Processing Association Annual Summit and Conference 2021 (APSIPA)*, December 2021.
- **Yosuke Higuchi**, Nanxin Chen, Yuya Fujita, Hirofumi Inaguma, Tatsuya Komatsu, Jaesong Lee, Jumon Nozaki, Tianzi Wang, Shinji Watanabe, “A Comparative Study on Non-Autoregressive Modelings for Speech-to-Text Generation,” In *Proceedings of 2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, December 2021.
- **Yosuke Higuchi**, Niko Moritz, Jonathan Le Roux, Takaaki Hori, “Momentum Pseudo-Labeling for Semi-Supervised Speech Recognition,” In *Proceedings of 22nd Annual Conference of International Speech Communication Association (INTERSPEECH)*, August 2021.
- Pengcheng Guo, Florian Boyer, Xuankai Chang, Tomoki Hayashi, **Yosuke Higuchi**, Hirofumi Inaguma, Naoyuki Kamo, Chenda Li, Daniel Garcia-Romero, Jiatong Shi, Jing Shi, Shinji Watanabe, Kun Wei, Wangyou Zhang, Yuekai Zhang, “Recent Developments on ESPnet Toolkit Boosted by Conformer,” In *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2021.
- Hirofumi Inaguma, **Yosuke Higuchi**, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, “Orthros: Non-autoregressive End-to-end Speech Translation with Dual-decoder,” In *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2021.
- **Yosuke Higuchi**, Shinji Watanabe, Hirofumi Inaguma, Tetsuji Ogawa, Tetsunori Kobayashi, “Improved Mask-CTC for Non-Autoregressive End-to-End ASR,” In *2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, June 2021.
- Shinji Watanabe, Florian Boyer, Xuankai Chang, Pengcheng Guo, Tomoki Hayashi, **Yosuke Higuchi**, Takaaki Hori, Wen-Chin Huang, Hirofumi Inaguma, Naoyuki Kamo, Shigeki Karita, Chenda Li, Jing Shi, Aswin Shanmugam Subramanian, Wangyou Zhang “The 2020 ESPnet Update: New Features, Broadened Applications, Performance Improvements, and Future Plans,” In *Proceedings of 2021 IEEE Data Science & Learning Workshop (DSLW)*, June 2021.
- **Yosuke Higuchi**, Naohiro Tawara, Tomoharu Iwata, Atsunori Ogawa, Tetsunori Kobayashi, Tetsuji Ogawa, “Noise-robust Attention Learning for End-to-End Speech Recognition,” In *Proceedings of 28th European Signal Processing Conference (EUSIPCO)*, January 2021.
- **Yosuke Higuchi**, Shinji Watanabe, Nanxin Chen, Tetsuji Ogawa, Tetsunori Kobayashi, “Mask CTC: Non-Autoregressive End-to-End ASR with CTC and Mask Predict,” In *Proceedings of 21th Annual Conference of International Speech Communication Association (INTERSPEECH)*, October 2020. (to appear)

- **Yosuke Higuchi**, Masayuki Suzuki, Gakuto Kurata, “Speaker Embeddings Incorporating Acoustic Conditions for Diarization,” In *Proceedings of 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, May 2020. (to appear)
- **Yosuke Higuchi**, Naohiro Tawara, Tetsunori Kobayashi, Tetsuji Ogawa, “Speaker Adversarial Training of DPGMM-based Feature Extractor for Zero-resource Languages,” In *Proceedings of 20th Annual Conference of International Speech Communication Association (INTERSPEECH)*, September 2019.

Preprint

- **Yosuke Higuchi**, Tetsuji Ogawa, Tetsunori Kobayashi, Shinji Watanabe, “BECTRA: Transducer-based End-to-End ASR with BERT-Enhanced Encoder,” In *arXiv preprint arXiv:2211.00792*, November 2022.
- **Yosuke Higuchi**, Tetsuji Ogawa, Tetsunori Kobayashi, Shinji Watanabe, “InterMPL: Momentum Pseudo-Labeling with Intermediate CTC Loss,” In *arXiv preprint arXiv:2211.00795*, November 2022.
- Brian Yan, Siddharth Dalmia, **Yosuke Higuchi**, Graham Neubig, Florian Metze, Alan W Black, Shinji Watanabe, “CTC Alignments Improve Autoregressive Translation,” In *arXiv preprint arXiv:2210.05200*, October 2022.
- Hirofumi Inaguma, **Yosuke Higuchi**, Kevin Duh, Tatsuya Kawahara, Shinji Watanabe, “Non-autoregressive End-to-end Speech Translation with Parallel Autoregressive Rescoring,” In *arXiv preprint arXiv:2109.04411*, September 2021.

Awards

Yahoo! JAPAN Award

from Information Processing Society of Japan (IPSJ) SIG-SLP

January 2020

Yamashita SIG Research Award

from Information Processing Society of Japan (IPSJ)

August 2020

Best Student Presentation Award

from the Acoustical Society of Japan (ASJ)

November 2020

ISS Young Researcher's Award in Speech Field

from the Institute of Electronics, Information and Communication Engineers (IEICE)

September 2021

Grants

Super Global University

from ICT & Robotics, Waseda University

December 2019 - March 2020

Repayment Exemption for Graduate Students with Excellent Achievements (Type I; full-exemption)

from Japan Student Services Organization (JASSO)

April 2019 - March 2021

Research Fellowship for Young Scientists (DC1)

from Japan Society for the Promotion of Science (JSPS)

April 2021 - March 2024

ACT-X Frontier of mathematics and information science

from Japan Science and Technology Agency (JST)

October 2021 - March 2024

Super Global University

from ICT & Robotics, Waseda University

March 2022 - June 2022

Skills

Natural language Japanese (native), English (fluent)

Programming language Python, C, C++, Java, JavaScript, LaTeX

Software & Tools PyTorch, Chainer, ESPnet, Kaldi